Yumeng He

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RESEARCH INTERESTS

My research interests are physical AI, robotics, computer vision, and graphics, with a focus on object/scene generation, articulated and kinematic asset modeling, physics-based simulation, real-to-sim pipelines, and policy learning.

EDUCATION

University of California, Los Angeles — Sep 2026 – Present
Ph.D. in Computer Science — *Los Angeles, CA, USA*
- Advisors: Prof. Chenfanfu Jiang and Prof. Demetri Terzopoulos

University of Southern California — Aug 2024 – May 2026
M.S. in Computer Science (Thesis) — *Los Angeles, CA, USA*
- Advisor: Prof. Jernej Barbič

University of Toronto St. George Campus — Sep 2019 – May 2024
Honours Bachelor of Science with High Distinction — *Toronto, ON, Canada*
- Computer Science and Mathematics double major

PUBLICATIONS AND MANUSCRIPTS [Google Scholar]

* Equal Contribution

Peer-reviewed Publications

[1] **One Spool at a Time: Fabricable Multicolor String Art**
Qiyang Chen, **Yumeng He**, Jiayin Lu, Yin Yang, Ying Jiang, Chenfanfu Jiang
SIGGRAPH Asia Poster, 2026

[2] **Beyond Kirchhoff–Love: A Volumetric Approach to Thin-Shell Mechanics**
Huanyu Chen, **Yumeng He**, Jernej Barbic
ACM Transactions on Graphics (SIGGRAPH Asia), 2026

[3] **Vibrational Affine Body Dynamics**
Haoyang Shi, **Yumeng He**, Sheldon Andrews, Chenfanfu Jiang, Kui Wu, Yin Yang
ACM Transactions on Graphics (SIGGRAPH Asia), 2026

[4] **SeeClear: Reliable Transparent Object Depth Estimation via Generative Opacification**
Xiaoying Wang*, **Yumeng He***, Jingkai Shi*, Jiayin Lu, Yin Yang, Ying Jiang, Chenfanfu Jiang
European Conference on Computer Vision (ECCV), 2026

[5] **SPARK: Sim-ready Part-level Articulated Reconstruction with VLM Knowledge**
Yumeng He*, Ying Jiang*, Jiayin Lu*, Yin Yang, Chenfanfu Jiang
Computer Vision and Pattern Recognition (CVPR), 2026 Oral (top 3.45%), CVPR Compute Transparency Champion

[6] **Birth of a Painting: Differentiable Brushstroke Reconstruction**
Ying Jiang*, Jiayin Lu*, Yunuo Chen*, **Yumeng He**, Kui Wu, Yin Yang, Chenfanfu Jiang
Conditionally Accepted by ACM Transactions on Graphics (SIGGRAPH), 2026

[7] **Self-supervised dual-layer 2D normalizing flow method for industrial anomaly detection**
Zhenlian Miao, Guangzhu Chen, Xiaojuan Liao, Jiu Dai, **Yumeng He**
Applied Soft Computing (ASOC), 2024.

Preprints

[8] **ArticulateArena: A Metric for Articulated Kinematics**
Yumeng He, Yongfei She, Huanyu Chen, Chun Yuan, Peihao Li, Joseph Masterjohn, Yin Yang, Ying Jiang, Chenfanfu Jiang
arXiv, 2026

[9] **Fishbone: From One 3D Asset to a Million Controllable Edits**
Yumeng He, Xiaoying Wang, Peihao Li, Yanjia Huang, Joe Masterjohn, Jiajun Wu, Leonidas Guibas, Yin Yang, Ying Jiang, Chenfanfu Jiang
arXiv, 2026

[10] **VoroLight: Learning Voronoi Surface Meshes via Sphere Intersection**
Jiayin Lu*, Ying Jiang*, **Yumeng He***, Yin Yang, Chenfanfu Jiang
arXiv, 2025

Thesis

[11] **From Kirchhoff–Love Shells to Volumetrically Consistent Shells: Understanding the Limitations of Classical Thin-Shell Theory and Developing the Two-Mode VC Shells**
Yumeng He, Jernej Barbic
Master's Thesis, University of Southern California, 2026.

RESEARCH EXPERIENCE

UCLA AIVC Lab | Visiting Student — Jun 2025 – Jul 2026
Supervisor: Prof. Chenfanfu Jiang — *Los Angeles, CA, USA*

USC Graphics | Master Thesis — Aug 2025 – May 2026
Supervisor: Prof. Jernej Barbič — *Los Angeles, CA, USA*

USC RESL Lab | Research Assistant — Jul 2025 – Dec 2025
Supervisor: Prof. Gaurav S. Sukhatme — *Los Angeles, CA, USA*

INDUSTRY EXPERIENCE

HCL Canada Inc. — Aug 2022 – Aug 2023
Software Developer Intern | *Full Time* — *Toronto, ON, Canada*

ACADEMIC SERVICE

- Reviewer: SIGGRAPH Asia (2026), Pacific Graphics (2026), TVCG (2026), CAVW (2026), ICLR (2027)
- Special Session Organizer, ACM SIGCSE — 2026
- Workshop Volunteer, SIGGRAPH — 2026
- Mentor for Viterbi Graduate Mentorship Program — Fall 2025
- Mentor for Women in Engineering Mentorship — Fall 2025
- Mentor for Viterbi Graduate Mentorship Program — Summer 2025

INVITED TALKS

- *MPC Is All You Need.* [Co-presented]. USC SLURM Lab, hosted by Prof. Daniel Seita — Aug 2025

AWARDS & HONORS

- CVPR Compute Transparency Champion (SPARK) — 2026
- Oral Presentation (top 3.45%), CVPR (SPARK) — 2026
- Dean's List Scholar — 2020 - 2024

MENTORING

- Xuanyu Pan, UCLA M.S. Student — Oct 2026 – Present
- Yukun Zhou, USC M.S. Student — Sep 2026 – Present
- Yongfei She, USC M.S. Student — May 2026 – Oct 2026
- Joyce Wang, CMU Undergraduate Student — Apr 2026 – Aug 2026
- Yuhan Zhou, UCLA Undergraduate Student — Apr 2026 – Jul 2026

TECHNICAL SKILLS

- **Programming**: Python, C++, Bash, GLSL
- **Graphics & Scientific Computing**: OpenGL, Eigen
- **Development Tools**: Linux, Git, Docker, GitHub Actions